242908: tuffaceous volcanic rock, Red Hill mine

(Ashburton Formation, Wyloo Group, Ashburton Basin, Capricorn Orogen)

Location and sampling

Yarraloola (SF 50-6), Pannawonica (2154)
MGA Zone 50, 404362E 7579101N

Sampled on 18 September 2019

This sample was collected from outcrop on Red Hill Station, about 9.4 km north of Red Hill homestead, 9.0 km south-southeast of Lassie Well, and 3.0 km southwest of the historic Red Hill mine.

Geological context

The unit sampled is a felsic volcanic rock of the 2008–1799 Ma Ashburton Formation of the Wyloo Group (Thorne and Seymour, 1991; Thorne and Johnson, 2020a) in the Ashburton Basin. The Ashburton Formation is about 6.5 km thick, and consists of greenschist facies mudstone, siltstone and lithic quartz sandstone, together with minor conglomerate, felsic to mafic volcanic rocks, banded iron-formation and chert, dolostone, and quartzite. The formation conformably overlies the Duck Creek Dolomite and grades laterally into the June Hill Volcanics (Evans et al., 2003; Wilson et al., 2010; Thorne and Johnson, 2020b). Both the Ashburton Formation and June Hill Volcanics are unconformably overlain by Paleoproterozoic siliciclastic sedimentary rocks of the Blair, Mount Minnie, Bresnahan, and Edmund Basins and by Mesozoic rocks of the Carnarvon Basin (Thorne and Johnson, 2020a). The present sample was collected close to the basal contact with the underlying Duck Creek Dolomite.

Petrographic description

The sample is a coarse-grained tuffaceous rock (Fig. 1) of probable rhyolitic composition, and consists of about 70% matrix, 15–18% plagioclase, 6–7% quartz, 5% quartz-replaced laths, 2% lithic grains, 1% chlorite-replaced grains, and accessory zircon. The matrix is composed of clay- or chlorite-altered microcrystalline material, and contains abundant non-welded relict glass shards, which have been replaced by fine chalcedonic quartz. The shard shapes are well preserved and include bubble wall, cuspate and rod-like forms. There is no evidence of welding of the shards. Plagioclase forms tabular phenocrysts up to 1.2 mm long and smaller, angular crystal fragments. Plagioclase has been completely replaced by sericite, and only a few grains exhibit faint twinning. Quartz forms anhedral, embayed, fractured phenocrysts up to 1.5 mm in size, and more abundant smaller angular fragments. Lath-shaped crystals, replaced by fine chalcedonic quartz and tiny magnetite needles, range up to 2.5 mm in length, and exhibit a laminated texture owing to secondary magnetite inclusions. It is possible these laths were originally small pumice clasts, although their very tabular forms and abruptly terminated ends, as well as the laminar texture, is more consistent with the laths representing replaced biotite. The chloritized grains also contain fine-grained secondary magnetite but are equant in shape, up to 0.5 mm in size and are most likely a replaced ferromagnesian mineral. Probable lithic grains are variably clay and limonite altered, up to 1.5 mm in size, and have a feathery feldspathic (?albite) texture. These could be clasts of altered coherent lava and crystal-vitric tuff with a similar composition as the host rock. There is no evidence of reworking in this sample, and it is interpreted as a tuff of primary pyroclastic origin. There is no evidence of deformation or metamorphism.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mainly subhedral to euhedral. The crystals are up to 350 μm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, all zircons exhibit concentric zoning, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11–12 February 2021, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.23% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-two analyses were obtained from 19 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Figure 1. Outcrop image for sample 242908: tuffaceous volcanic rock, Red Hill mine

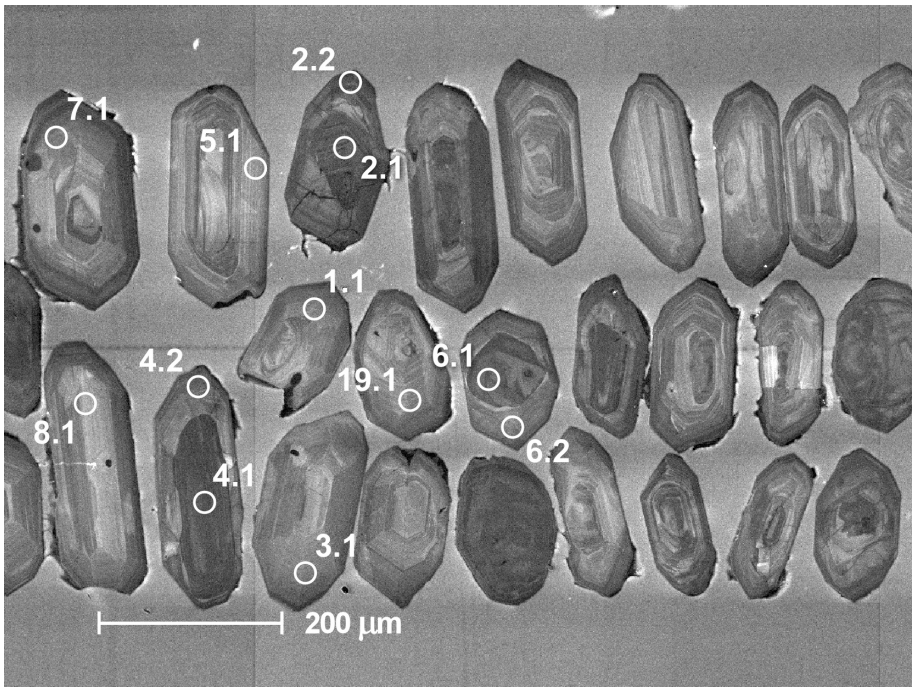

Figure 2. Cathodoluminescence image of representative zircons from sample 242908: tuffaceous volcanic rock, Red Hill mine. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 242908: tuffaceous volcanic rock, Red Hill mine

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	13	10.1	284	75	0.27	0.130	3.158	0.028	0.10988	0.00067	3.162	0.028	0.10876	0.00076	1771	14	1779	13	0.4
I	12	9.1	158	87	0.55	0.112	3.115	0.033	0.11022	0.00088	3.119	0.033	0.10924	0.00098	1793	17	1787	16	-0.3
I	20	17.1	177	76	0.43	0.040	3.117	0.031	0.10959	0.00082	3.119	0.031	0.10924	0.00086	1793	16	1787	14	-0.3
I	17	14.1	186	66	0.35	0.000	3.180	0.032	0.10930	0.00080	3.180	0.032	0.10930	0.00080	1763	15	1788	13	1.4
I	11	8.1	125	58	0.46	0.029	3.132	0.036	0.10971	0.00097	3.133	0.036	0.10946	0.00101	1786	18	1790	17	0.3
I	9	6.2	162	34	0.21	0.000	3.195	0.033	0.10961	0.00083	3.195	0.033	0.10961	0.00083	1755	16	1793	14	2.1
I	16	13.1	247	58	0.24	0.318	3.134	0.029	0.11261	0.00071	3.144	0.029	0.10985	0.00093	1780	14	1797	15	0.9
I	15	12.1	211	49	0.23	0.017	3.118	0.030	0.11012	0.00077	3.119	0.030	0.10998	0.00078	1793	15	1799	13	0.3
I	1	1.1	187	82	0.44	-0.020	3.183	0.032	0.10989	0.00083	3.182	0.032	0.11007	0.00085	1762	16	1801	14	2.2
I	10	7.1	191	72	0.38	-0.038	3.127	0.031	0.10976	0.00080	3.126	0.031	0.11009	0.00084	1789	16	1801	14	0.7
I	3	2.2	349	86	0.25	0.095	3.140	0.027	0.11093	0.00062	3.143	0.027	0.11010	0.00068	1781	13	1801	11	1.1
I	18	15.1	194	85	0.44	-0.073	3.137	0.031	0.10950	0.00078	3.135	0.031	0.11014	0.00084	1785	15	1802	14	0.9
I	7	5.1	128	47	0.37	0.057	3.104	0.035	0.11065	0.00097	3.106	0.035	0.11015	0.00104	1799	18	1802	17	0.1
I	19	16.1	108	55	0.51	-0.067	3.189	0.038	0.10984	0.00105	3.187	0.038	0.11042	0.00113	1759	19	1806	19	2.6
I	21	18.1	189	92	0.49	-0.077	3.170	0.031	0.10979	0.00081	3.167	0.031	0.11046	0.00088	1769	15	1807	14	2.1
I	4	3.1	182	81	0.45	0.000	3.215	0.032	0.11066	0.00081	3.215	0.032	0.11066	0.00081	1746	15	1810	13	3.6
I	5	4.1	454	255	0.56	0.803	3.197	0.025	0.11837	0.00121	3.223	0.025	0.11138	0.00144	1742	12	1822	23	4.4
I	6	4.2	280	84	0.30	-0.094	3.153	0.028	0.11063	0.00069	3.150	0.028	0.11144	0.00075	1777	14	1823	12	2.5
I	14	11.1	126	62	0.50	-0.115	3.200	0.036	0.11065	0.00099	3.196	0.036	0.11164	0.00111	1755	18	1826	18	3.9
I	22	19.1	166	90	0.55	-0.065	3.133	0.032	0.11115	0.00086	3.131	0.032	0.11172	0.00092	1787	16	1828	15	2.2
X	2	2.1	827	231	0.28	0.100	2.776	0.020	0.12634	0.00041	2.779	0.020	0.12546	0.00045	1981	12	2035	6	2.7
X	8	6.1	455	202	0.44	0.347	2.177	0.017	0.16216	0.00058	2.185	0.017	0.15908	0.00071	2429	16	2446	8	0.7

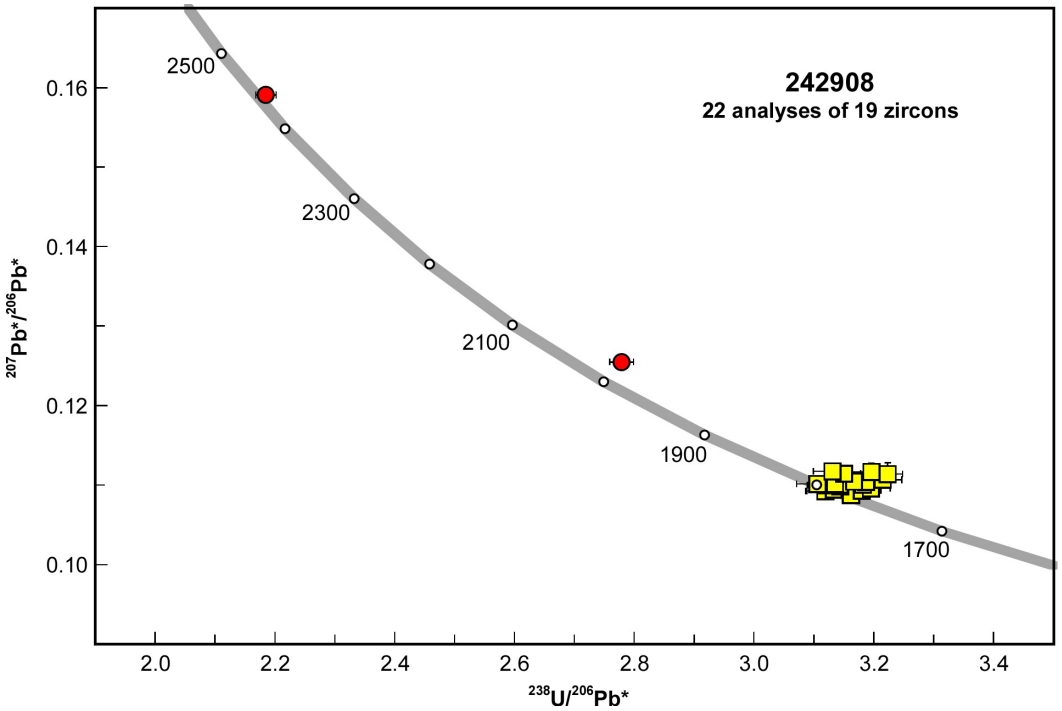

Figure 3. **U–Pb analytical data for zircons from sample 242908: tuffaceous volcanic rock, Red Hill mine. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores)**

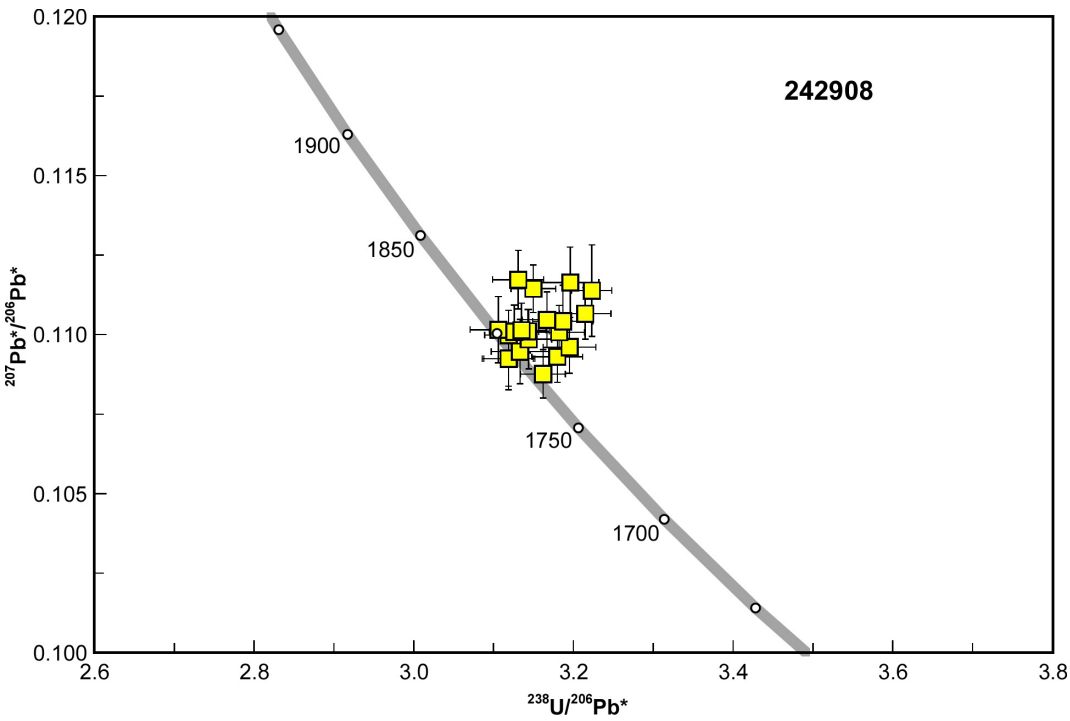

Figure 4. **Expanded view of U–Pb analytical data for zircons from sample 242908: tuffaceous volcanic rock, Red Hill mine. Symbols as in Figure 3**

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 20 analyses of 19 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 1801 ± 6 Ma (MSWD = 0.87).

Group X comprises two analyses of two zircon cores (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates (1σ) of 2446 ± 8 and 2035 ± 6 Ma.

The date of 1801 ± 6 Ma for the 20 analyses in Group I is interpreted as the igneous crystallization age. The dates (1σ) of 2446 ± 8 and 2035 ± 6 Ma for the two analyses in Group X are interpreted as the ages of inherited zircons.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Martin, DMcB 2021, 242908: tuffaceous volcanic rock, Red Hill mine; Geochronology Record 1819: Geological Survey of Western Australia, 5p.

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Data released: 12 November 2021